

PROJECT CHARTER

# AI-Powered CMO Response Time Reduction & Incident Management Automation System

Portfolio case study | Telecom Network Operations | Business Analysis, Agile Delivery and Six Sigma DMAIC

|                   |                                                                |
|-------------------|----------------------------------------------------------------|
| Project Sponsor   | Operations Leadership                                          |
| Project Lead      | Business Analyst / Project Lead / Automation Solution Designer |
| Primary Users     | Fiber Operations Team, Area Managers and NOC Team              |
| Delivery Approach | Agile Delivery + Six Sigma DMAIC                               |

## 1. Project Background

Telecom network operations relied on manual monitoring of incident notifications received from Network Operations Center (NOC) communication channels. Critical outage information was frequently shared in raw or unstructured formats, requiring manual filtering, identification of service-impacting incidents, stakeholder communication, follow-up tracking and escalation.

The absence of a standardized, automated incident-management workflow created delayed response, inconsistent ownership, manual reporting and limited operational visibility. The initiative was therefore designed as a single automation system combining reactive incident handling with proactive critical-incident monitoring and executive reporting.

## 2. Business Problem

- Raw incident notifications required manual review before the relevant operations team could act.
- Critical incidents could take approximately 10 to 45 minutes to reach the appropriate response team, depending on the alert and manual intervention required.
- Ownership assignment, follow-ups and escalation depended on dedicated operational resources.
- Critical fiber incidents were monitored through periodic data extraction and manual filtering, limiting proactive visibility.
- Leadership reporting was manual and did not provide a near real-time view of open incidents, performance or risk.
- These gaps increased response delays, weakened SLA governance and contributed to higher Cell Minute Outage (CMO) exposure.

### 3. Business Opportunity

Implement an AI-powered automation platform that standardizes the incident lifecycle from alert identification to closure and reporting. The solution will intelligently process alerts, assign ownership, automate follow-ups and escalations, monitor critical incidents proactively, and make operational KPIs available through dashboards.

### 4. Project Objective

Develop and implement an AI-powered incident management automation system that streamlines reactive and proactive incident handling, improves outage visibility, automates follow-up and escalation workflows, and optimizes CMO response time through intelligent automation and operational dashboards.

### 5. Project Goals

- Reduce CMO-related incident dockets and response delays.
- Improve near real-time incident visibility and operational decision-making.
- Automate repetitive monitoring, follow-up and reporting activities.
- Standardize incident communication, ownership and escalation.
- Strengthen SLA monitoring, accountability and operational governance.
- Provide actionable dashboards for operations leadership.

### 6. Project Scope

#### 6.1 In Scope

##### Module 1: Reactive Incident Automation

- Receive and process incident notifications from approved operational communication channels.
- Filter alerts and identify service-impacting or critical incidents.
- Generate structured alerts and automatically tag relevant stakeholders.
- Send scheduled follow-up reminders and escalate long-running incidents.
- Notify stakeholders on restoration and track the incident lifecycle.

##### Module 2: Proactive Incident Monitoring

- Capture and monitor critical incident updates.
- Produce area-wise pending-incident visibility and automated reminders.
- Publish executive dashboard views and operational KPI reports.
- Support proactive monitoring of open incidents and escalation status.

#### 6.2 Out of Scope

- Network fault resolution, field maintenance and hardware replacement.
- Core network configuration, capacity planning and physical infrastructure upgrades.
- Replacement of existing NOC systems or operational communication platforms.
- Automation of decisions that require authorized human approval.

## 7. Expected Business Benefits

| Operational Benefits                                    | Business Benefits                                         |
|---------------------------------------------------------|-----------------------------------------------------------|
| Faster identification and routing of critical incidents | Reduced CMO exposure and improved service availability    |
| Automated follow-ups, reminders and escalation          | Improved SLA adherence and accountability                 |
| Near real-time incident and KPI visibility              | Faster, data-driven operational decisions                 |
| Reduced manual monitoring and reporting effort          | Improved resource utilization and operational efficiency  |
| Standardized incident lifecycle communication           | Improved stakeholder coordination and customer experience |

## 8. Key Success Metrics

The following outcome comparison is based on the operational data provided for the implementation period. The primary outcome is a reduction in overall CMO-related incident dockets; category-level results are shown for transparency.

| KPI                                       | Before           | After           | Improvement / Outcome          |
|-------------------------------------------|------------------|-----------------|--------------------------------|
| <b>Total CMO-related incident dockets</b> | 607              | 494             | 19% reduction                  |
| <b>Dockets resolved in 30-60 minutes</b>  | 71               | 58              | 18% reduction                  |
| <b>Dockets resolved in 1-2 hours</b>      | 161              | 104             | 35% reduction                  |
| <b>Dockets open for more than 3 hours</b> | 248              | 202             | 19% reduction                  |
| <b>Incident visibility</b>                | Manual / delayed | Near real-time  | Improved operational awareness |
| <b>Follow-up tracking</b>                 | Manual           | Automated       | Standardized workflow          |
| <b>Leadership reporting</b>               | Manual           | Dashboard-based | Improved decision support      |

*Note: The solution is positioned as improving response, tracking and escalation effectiveness. It does not claim to prevent underlying network incidents unless separately evidenced.*

## 9. Deliverables

- Reactive Incident Automation module
- Proactive Incident Monitoring and Dashboard module
- Automated notification, reminder and escalation workflows
- Executive KPI dashboard and operational reports
- Business and functional documentation
- Test evidence, UAT support and deployment guidance

## 10. Stakeholders

| Stakeholder                   | Role / Interest                      | Key Responsibility                                                               |
|-------------------------------|--------------------------------------|----------------------------------------------------------------------------------|
| <b>Operations Leadership</b>  | Executive sponsor and business owner | Provide strategic direction; review outcomes and KPI performance                 |
| <b>Fiber Operations Team</b>  | Primary business users               | Respond to incidents; validate operational workflow and benefits                 |
| <b>NOC Team</b>               | Incident source                      | Provide incident notifications and operational inputs                            |
| <b>Area Managers</b>          | Response owners                      | Coordinate action, updates and local escalation                                  |
| <b>Circle Operations Head</b> | Escalation authority                 | Resolve critical exceptions and review operational performance                   |
| <b>Project Lead</b>           | Delivery owner                       | Analyse requirements, design solution, coordinate validation and manage delivery |

## 11. Assumptions

- Approved operational communication channels continue to provide incident notifications.
- Stakeholders use the structured automated alerts to support operational decision-making.
- Required access, infrastructure and support are available for deployment and maintenance.
- Existing operating processes remain in place except where automation improves monitoring, follow-up and reporting.

## 12. Constraints

- Solution performance depends on the quality, timeliness and format of incoming incident notifications.
- The solution supports operational decision-making; it does not perform network fault resolution.
- Availability of communication platforms, source data and dashboards can affect end-to-end performance.
- Public portfolio materials must not contain company names, personal identifiers, group IDs, credentials, system addresses or confidential source data.

## 13. Risks and Mitigation

| Risk                                     | Potential Impact                                             | Mitigation                                                                       |
|------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Incoming message format changes</b>   | Parsing logic may not correctly classify or route an alert.  | Maintain configurable rules; validate samples and update parsing logic promptly. |
| <b>Communication platform downtime</b>   | Delayed delivery of alerts, reminders or escalation notices. | Monitor bot health; retain an approved manual fallback and retry mechanism.      |
| <b>Incorrect incident categorization</b> | Incorrect stakeholders may be notified, affecting response.  | Use validation checks, periodic sampling and iterative rule refinement.          |
| <b>User adoption</b>                     | Teams may continue to rely on                                | Provide onboarding, simple job aids and feedback-                                |

| Risk                            | Potential Impact                                   | Mitigation                                                                 |
|---------------------------------|----------------------------------------------------|----------------------------------------------------------------------------|
| challenges                      | manual follow-up processes.                        | led workflow refinement.                                                   |
| Dashboard or data refresh delay | Leadership may not see the latest incident status. | Set refresh monitoring and clearly communicate data currency in reporting. |

## 14. Project Methodology

| Discipline        | Application to this Project                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Business Analysis | Identify business problem, stakeholders, requirements, scope, processes, KPIs and acceptance criteria.                                                      |
| Agile Delivery    | Deliver and validate the solution iteratively with operational users; incorporate feedback into successive releases.                                        |
| Six Sigma DMAIC   | Define the response-time problem; measure baseline performance; analyze root causes; improve through automation; control through dashboards and monitoring. |
| Solution Design   | Design secure, sanitized automation workflows for alert processing, notification, escalation, tracking and reporting.                                       |
| Testing and UAT   | Execute functional testing and user acceptance testing against documented requirements and real-world scenarios.                                            |

## 15. Charter Approval

This charter provides the agreed business context, scope and intended outcomes for the portfolio case study. Names, systems and organization-specific information have been anonymized for public sharing.

| Approver Role                           | Name / Signature | Date  |
|-----------------------------------------|------------------|-------|
| Project Sponsor / Operations Leadership | _____<br>_____   | _____ |
| Project Lead                            | _____<br>_____   | _____ |
| Primary Business Representative         | _____<br>_____   | _____ |